

QEM-8³: Entanglement Square Lattice and Flip-Wave Dynamics

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Abstract

This work extends the QEM-8 framework from two-dimensional pair modes to a three-dimensional lattice representation. The entanglement relations AB, BC, CD, DA form square oppositions. When layered into 3D, stable configurations must preserve absolute duality across all axes. We derive the allowed 3D configurations and show that collapse propagation behaves as discrete flip-wave transitions across the lattice.

Main Discussion

1. QEM-8 Base Structure

We begin with the eight directional pair-modes AB, BA, CD, DC, AC, CA, BD, DB.

2. 3D Extension

Stacking layers introduces vertical dual constraints. Only configurations preserving full duality remain.

This results in 32 valid 3D entanglement lattice patterns.

3. Flip-Wave Collapse

Collapse propagates as a wave of relational flips maintaining dual symmetry.

4. Significance

The system suggests entanglement may encode intrinsic relational topology independent of space-time.